

Credit-Market Sentiment and the Business Cycle: A Replication of López-Salido et al. (2017)

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Abstract

We replicate the core results of López-Salido et al. (2017), who show that elevated credit-market sentiment forecasts a slowdown in real activity about two years ahead. We describe the data, present summary statistics of the underlying series, and reproduce every table and figure across the replication sample, an extended sample, and an Aaa-spread variant.

1 Overview

When credit is “frothy”—the López-Salido et al. (2017) thesis—spreads are narrow and low-quality issuance is high; those conditions mean-revert as activity turns down.

2 Data Sources

- **FRED** (Federal Reserve Bank of St. Louis): Baa/Aaa yields, Treasury yields, GDP. Several series are a newer vintage and must be stitched together across multiple FRED series; where the appendix is silent about which underlying series to use, we made documented judgement calls.
- **Shiller** (Shiller, 2015): long-run market data and the P/E10 ratio.
- **Greenwood–Hanson high-yield share** (Greenwood and Hanson, 2013), pulled from the HBS workbook (Harvard Business School Behavioral Finance and Financial Stability Project) and spliced with Mergent FISD; a newer vintage than the paper.

3 Summary Statistics

The following tables and figures characterize the underlying series before the regression results, so the reader can gauge scale, dispersion, and co-movement of the data.

Table 1: Summary statistics: cape.

<i>Shiller Annual Stock-Market Series, 1929–2025 (N = 97)</i>								
Variable	Units	Mean	Std. Dev.	Min	Q1	Median	Q3	Max
S&P Composite price index	index level	758.66	1,318.14	6.82	26.04	106.50	1,110.38	6,853.03
S&P Composite dividend	\$ per share	13.83	19.10	0.44	1.47	4.67	16.27	79.52
Cyclically adj. P/E (CAPE)	ratio	19.23	8.48	7.83	11.75	17.56	24.86	44.20
S&P total return (log)	pct.	9.54	18.06	-51.58	0.23	14.26	21.11	42.29

Table 2: Summary statistics: credit spreads.

<i>FRED Monthly Series, 1925:01–2025:12 (N = 1,212)</i>								
Variable	Units	Mean	Std. Dev.	Min	Q1	Median	Q3	Max
Baa-Treasury spread	pp.	1.98	0.95	0.29	1.23	1.92	2.48	7.24
Aaa-Treasury spread	pp.	0.88	0.54	-0.17	0.39	0.81	1.23	2.68
Baa corporate bond yield	pct.	6.75	2.87	2.94	4.80	6.01	8.34	17.18
Aaa corporate bond yield	pct.	5.64	2.71	2.14	3.57	4.74	7.36	15.49
10-year Treasury yield	pct.	4.77	2.75	0.62	2.65	3.92	6.29	15.32
Months in NBER recession	pct. of months	16.6	37.2	0.0	0.0	0.0	0.0	100.0

Table 3: Summary statistics: gdp growth.

<i>FRED Annual Series, 1929–2024 (N = 96)</i>								
Variable	Units	Mean	Std. Dev.	Min	Q1	Median	Q3	Max
Real GDP	\$tn	8.53	6.58	0.88	2.85	6.54	14.13	23.36
Population	millions	222.5	68.9	121.7	158.9	219.2	283.1	340.1
Real GDP per capita	\$000s	32.9	17.8	7.0	18.1	29.8	49.9	68.7
CPI inflation (Dec/Dec)	pct.	3.04	3.79	-10.84	1.49	2.76	4.08	16.66
Baa corporate bond yield	pct.	6.83	2.93	3.10	4.79	6.17	8.44	16.55
Aaa corporate bond yield	pct.	5.66	2.74	2.26	3.49	4.71	7.33	14.23
10-year Treasury yield	pct.	4.78	2.73	0.93	2.51	4.03	6.31	13.72
3-month Treasury yield	pct.	3.46	3.21	0.01	0.39	3.00	5.29	16.35
Baa-Treasury spread	pp.	2.05	1.07	0.40	1.25	1.92	2.56	6.49
Aaa-Treasury spread	pp.	0.88	0.55	-0.11	0.39	0.80	1.25	2.63

Table 4: Summary statistics: hy share.

<i>Greenwood-Hanson High-Yield Share, 1926–2026 (N = 101)</i>								
Variable	Units	Mean	Std. Dev.	Min	Q1	Median	Q3	Max
High-yield issuance share	pct.	19.4	14.8	0.0	7.0	15.1	28.6	63.9
Log high-yield share	log	-2.05	1.11	-6.02	-2.61	-1.86	-1.25	-0.45

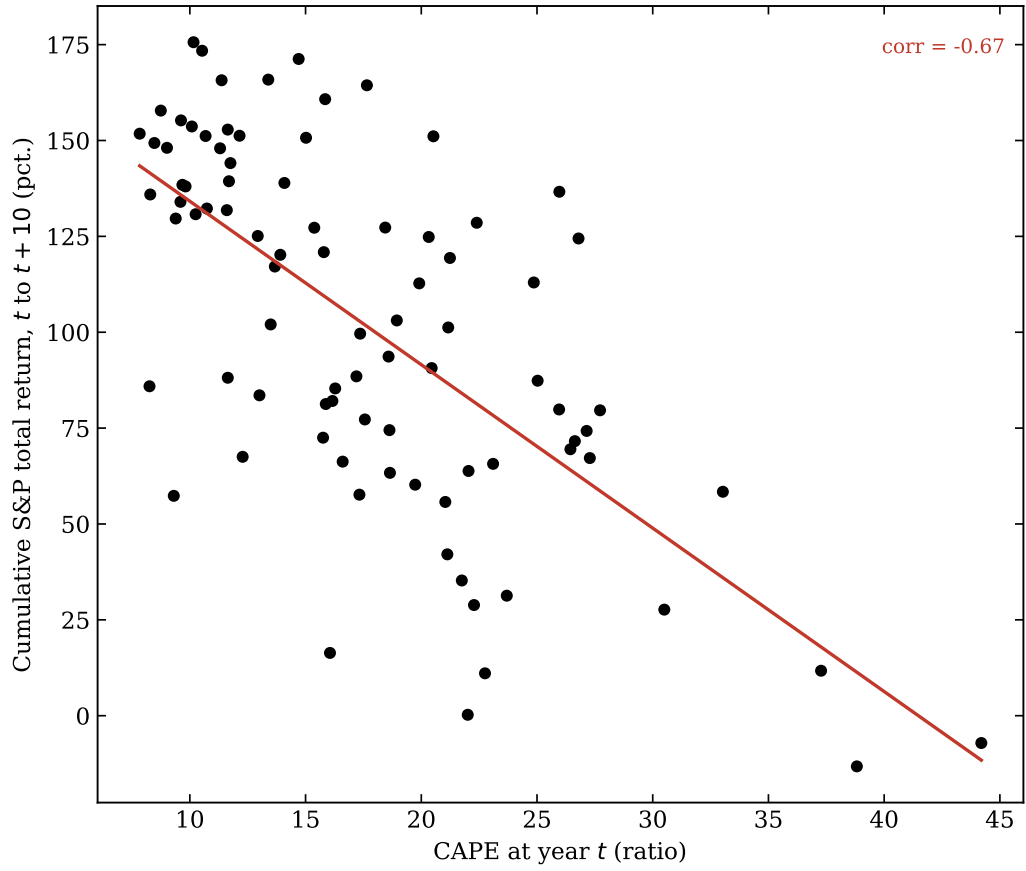


Figure 1: Summary statistics: cape.

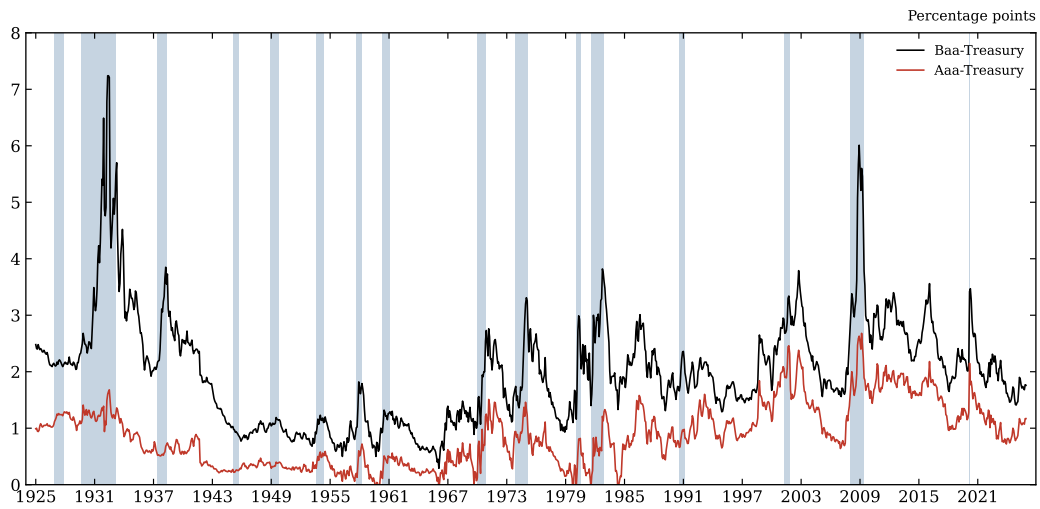


Figure 2: Summary statistics: credit spreads.

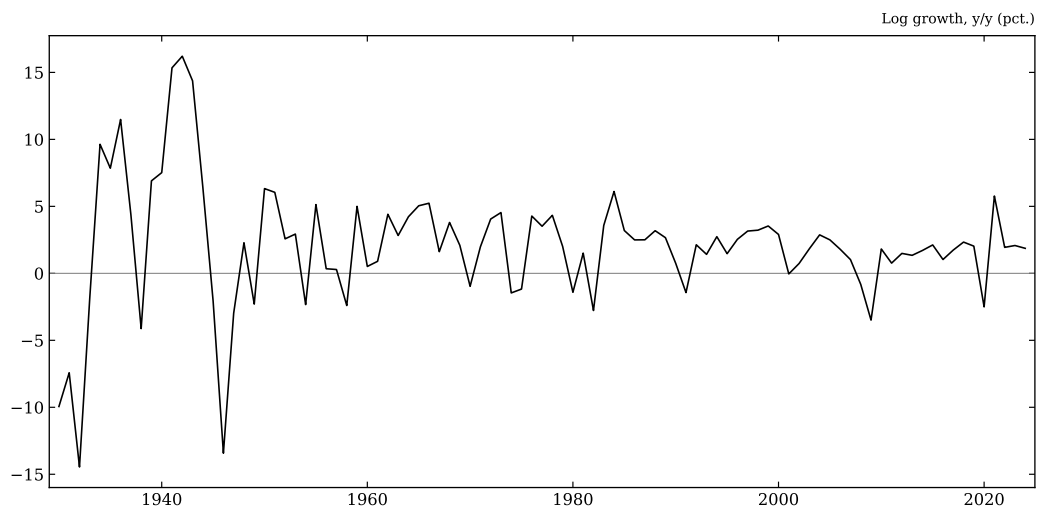


Figure 3: Summary statistics: gdp growth.

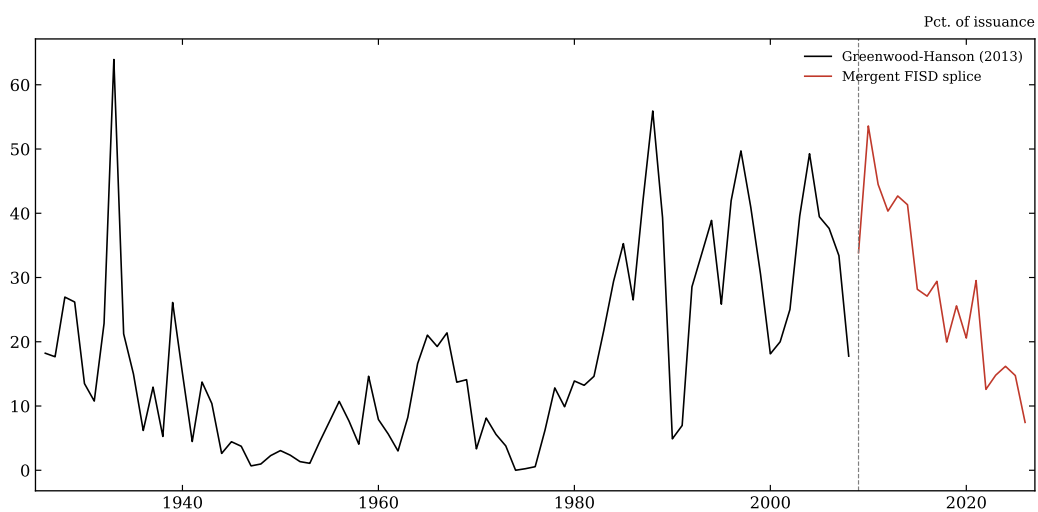


Figure 4: Summary statistics: hy share.

4 Replication Sample (1929–2015)

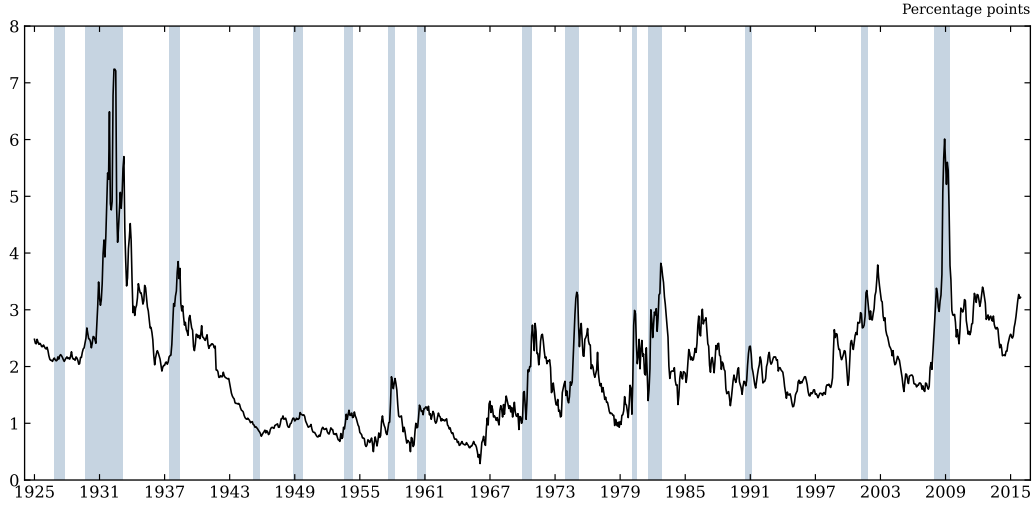


Figure 5: Figure I. Baa–Treasury credit spread with NBER recessions shaded. *Takeaway:* the spread spikes into every recession, motivating it as a cyclical signal.

Table 5: Table I. One-year-ahead GDP growth on the lagged credit-spread change vs. equity returns. *Takeaway:* the credit term is significant where equity returns are not.

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.958*** (0.585)	—	-2.124*** (0.578)
r_t^{SP}	—	0.075** (0.033)	0.022 (0.035)
Δy_{t-1}	0.476*** (0.104)	0.478*** (0.136)	0.464*** (0.114)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.248 (0.290)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.814** (0.360)
π_{t-1}	—	—	0.095 (0.076)
$\bar{R}^2$	0.426	0.381	0.453
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.365	—	-0.396
r_t^{SP}	—	0.299	0.087

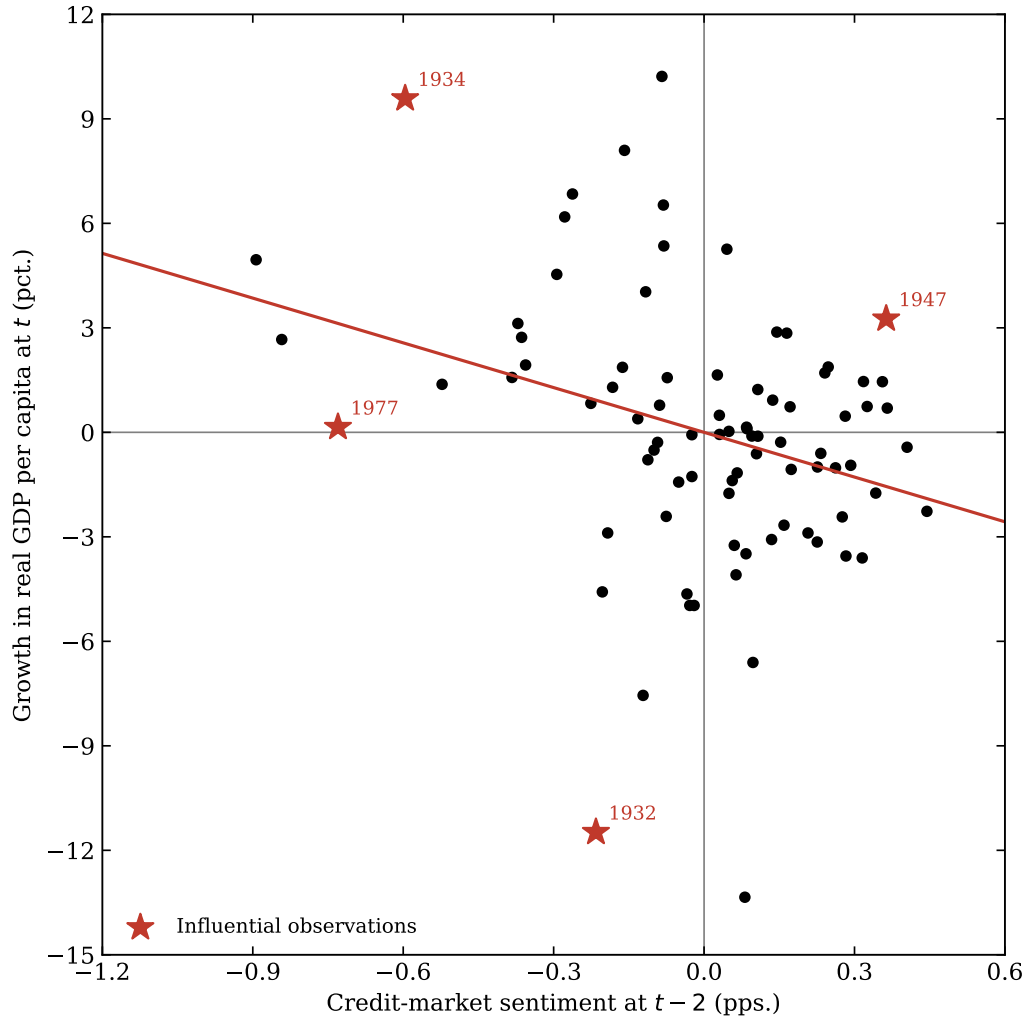


Figure 6: Figure II. Credit-market sentiment vs. subsequent growth. *Takeaway:* the downward slope shows frothy credit precedes weaker growth.

Table 6: Table II. Sentiment-based predictive regressions with the auxiliary decomposition. *Takeaway:* lagged log high-yield share and spread jointly forecast spread changes.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-4.281*** (1.253)	—	-3.861*** (1.301)	-4.907*** (1.678)
$\hat{r}_t^{SP}$	—	0.145* (0.082)	0.053 (0.074)	—
Δy_{t-1}	0.597*** (0.114)	0.534*** (0.099)	0.590*** (0.112)	0.581*** (0.096)
$\Delta i_{t-1}^{(3m)}$	—	—	—	0.099 (0.313)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.616 (0.414)
π_{t-1}	—	—	—	0.123 (0.154)
R^2	0.377	0.337	0.380	0.394
Auxiliary regressions		Δs_t	r_t^{SP}	
$\ln \text{HYS}_{t-2}$		0.147*** (0.042)	—	
s_{t-2}		-0.245*** (0.047)	—	
$\ln[P/E10]_{t-2}$		—	-0.132*** (0.040)	
R^2		0.102	0.083	

5 Extended Sample (1929–2025)

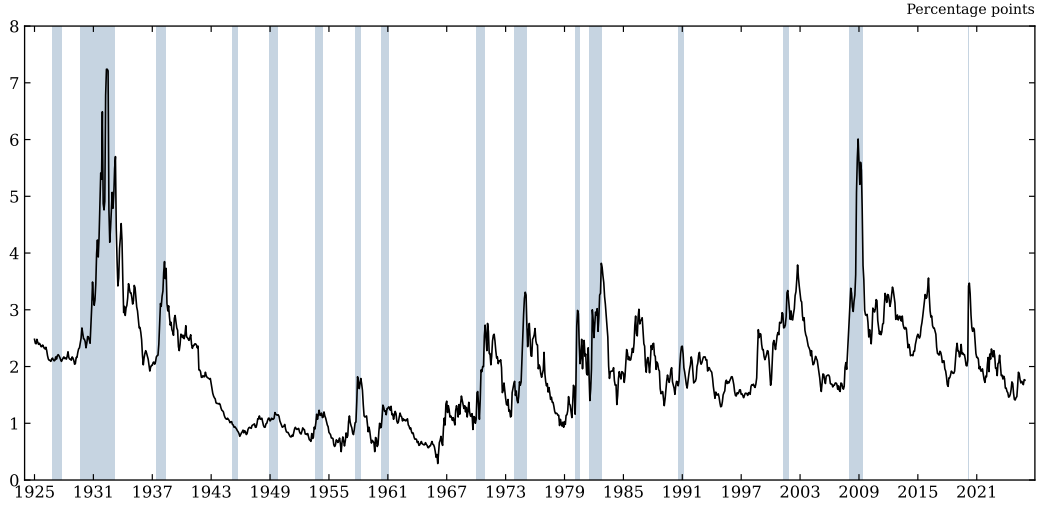


Figure 7: Figure I, extended. *Takeaway:* the cyclical pattern persists through recent data.

Table 7: Table I, extended sample. *Takeaway:* the credit result survives out of the original window.

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.874*** (0.566)	—	-2.020*** (0.572)
r_t^{SP}	—	0.070** (0.033)	0.018 (0.035)
Δy_{t-1}	0.461*** (0.108)	0.464*** (0.136)	0.450*** (0.116)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.183 (0.269)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.752** (0.367)
π_{t-1}	—	—	0.088 (0.078)
$\bar{R}^2$	0.395	0.354	0.415
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.350	—	-0.378
r_t^{SP}	—	0.286	0.072

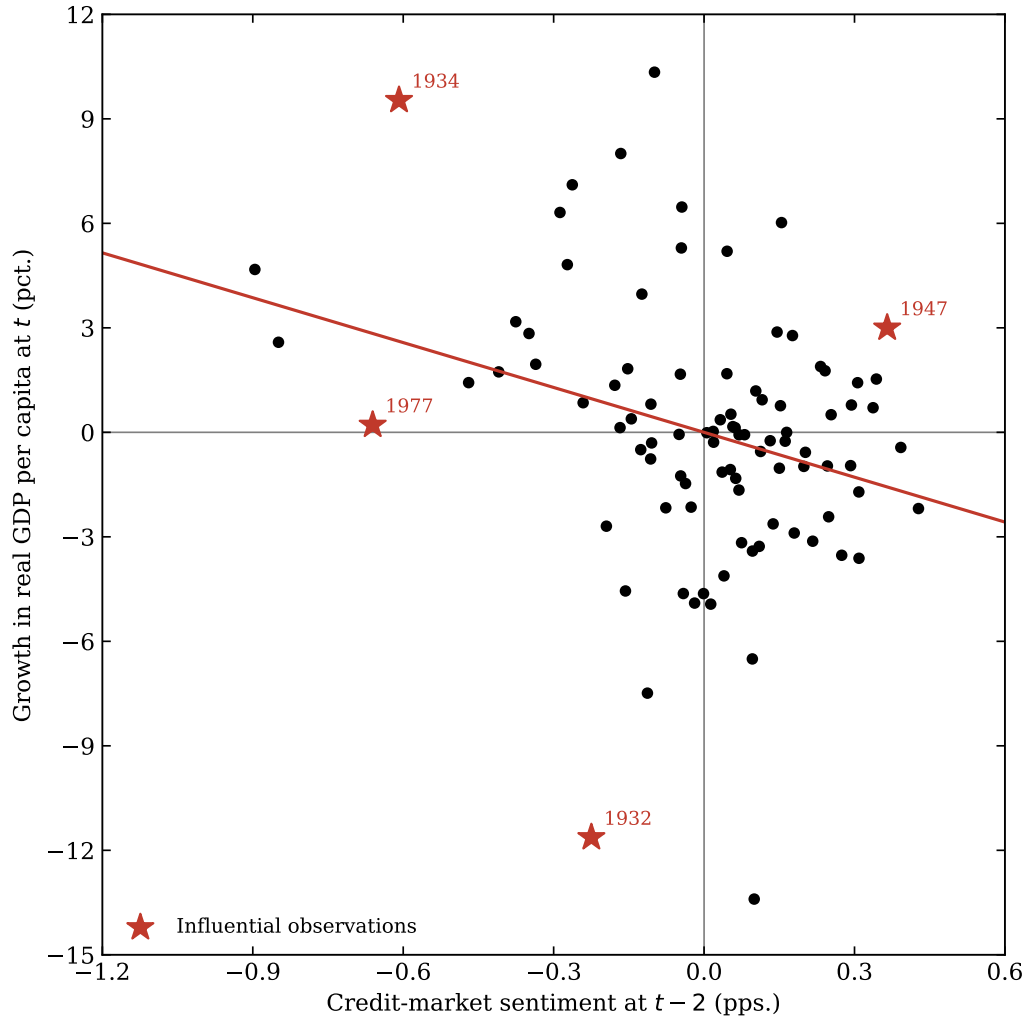


Figure 8: Figure II, extended. *Takeaway:* the negative sentiment–growth relation holds up.

Table 8: Table II, extended sample. *Takeaway:* coefficients are close to the replication values.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-4.294*** (1.295)	—	-3.901*** (1.316)	-4.943*** (1.703)
$\hat{r}_t^{SP}$	—	0.158* (0.092)	0.062 (0.081)	—
Δy_{t-1}	0.577*** (0.118)	0.516*** (0.103)	0.570*** (0.115)	0.559*** (0.100)
$\Delta i_{t-1}^{(3m)}$	—	—	—	0.123 (0.290)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.612 (0.410)
π_{t-1}	—	—	—	0.121 (0.154)
R^2	0.353	0.313	0.356	0.368
Auxiliary regressions				
		Δs_t	r_t^{SP}	
$\ln \text{HYS}_{t-2}$		0.129*** (0.039)	—	
s_{t-2}		-0.243*** (0.046)	—	
$\ln[P/E10]_{t-2}$		—	-0.103*** (0.034)	
R^2		0.095	0.059	

6 Extension: Aaa Credit Spread

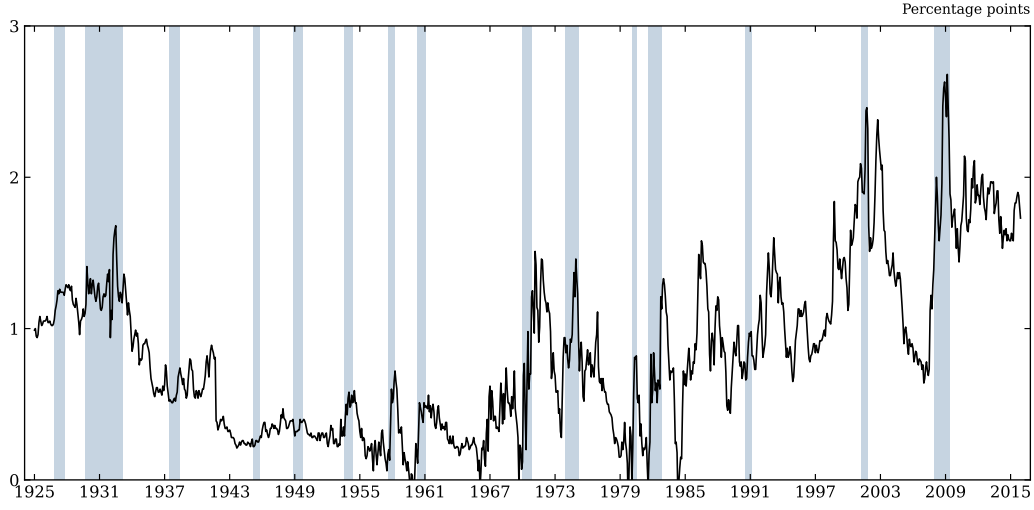


Figure 9: Figure I, Aaa (replication). *Takeaway:* the higher-grade spread is smoother but still cyclical.

Table 9: Table I, Aaa (replication). *Takeaway:* the Aaa spread carries weaker but same-signed predictive content.

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.958*** (0.734)	—	-3.192*** (1.168)
r_t^{SP}	—	0.075** (0.033)	0.056 (0.035)
Δy_{t-1}	0.517*** (0.109)	0.478*** (0.136)	0.477*** (0.121)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.326 (0.314)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.917* (0.504)
π_{t-1}	—	—	0.102 (0.108)
$\bar{R}^2$	0.321	0.381	0.396
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.165	—	-0.269
r_t^{SP}	—	0.299	0.225

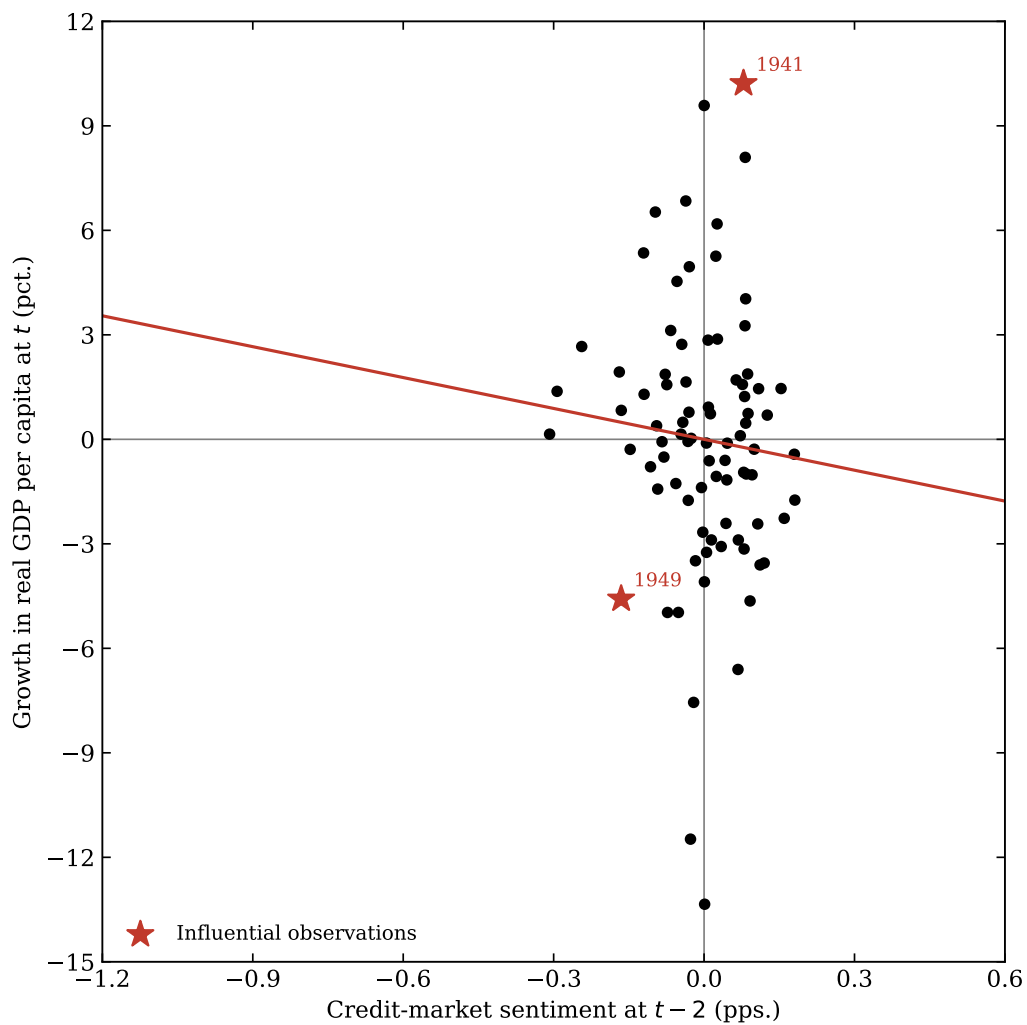


Figure 10: Figure II, Aaa (replication). *Takeaway:* sentiment built on Aaa shows a flatter slope.

Table 10: Table II, Aaa (replication). *Takeaway:* the auxiliary decomposition is weaker for the higher-grade spread.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-2.956 (2.488)	—	-2.362 (2.865)	-3.663 (2.806)
$\hat{r}_t^{SP}$	—	0.145* (0.082)	0.131 (0.085)	—
Δy_{t-1}	0.555*** (0.108)	0.534*** (0.099)	0.549*** (0.100)	0.559*** (0.105)
$\Delta i_{t-1}^{(3m)}$	—	—	—	-0.063 (0.308)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.373 (0.394)
π_{t-1}	—	—	—	0.043 (0.132)
R^2	0.321	0.337	0.342	0.328

Auxiliary regressions	Δs_t	r_t^{SP}
$\ln \text{HYS}_{t-2}$	0.078*** (0.029)	—
s_{t-2}	-0.167*** (0.058)	—
$\ln[P/E10]_{t-2}$	—	-0.132*** (0.040)
R^2	0.064	0.083

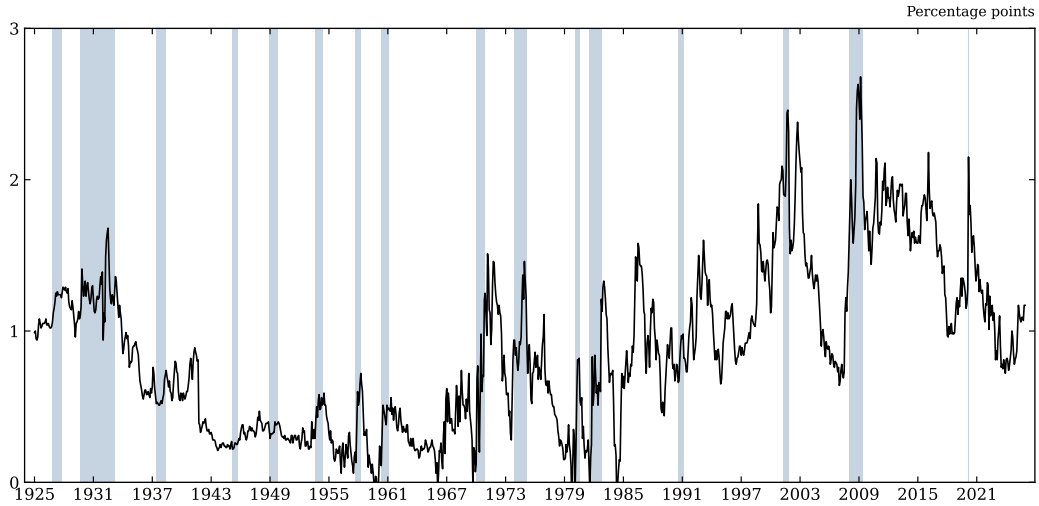


Figure 11: Figure I, Aaa (extended). *Takeaway:* the Aaa pattern extends to the present.

Table 11: Table I, Aaa (extended). *Takeaway:* extended Aaa estimates remain same-signed.

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.810** (0.714)	—	-2.873** (1.165)
r_t^{SP}	—	0.070** (0.033)	0.052 (0.035)
Δy_{t-1}	0.500*** (0.112)	0.464*** (0.136)	0.463*** (0.124)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.257 (0.289)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.797 (0.495)
π_{t-1}	—	—	0.091 (0.109)
$\bar{R}^2$	0.297	0.354	0.360
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.154	—	-0.244
r_t^{SP}	—	0.286	0.210

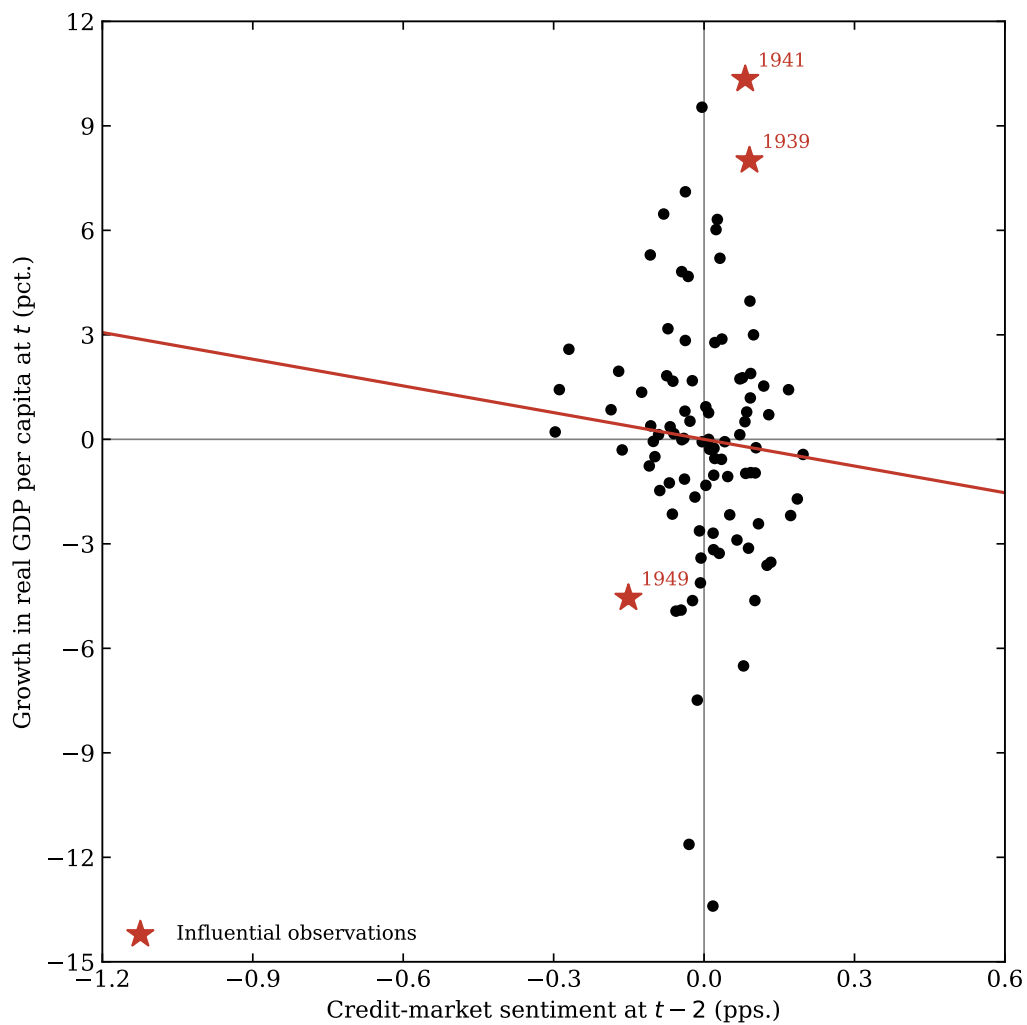


Figure 12: Figure II, Aaa (extended). *Takeaway:* the relation weakens but does not reverse.

Table 12: Table II, Aaa (extended). *Takeaway*: consistent with the replication-sample Aaa result.

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-2.556 (2.389)	—	-2.546 (2.578)	-3.219 (2.717)
$\hat{r}_t^{SP}$	—	0.158* (0.092)	0.146 (0.094)	—
Δy_{t-1}	0.536*** (0.111)	0.516*** (0.103)	0.532*** (0.103)	0.540*** (0.107)
$\Delta i_{t-1}^{(3m)}$	—	—	—	-0.050 (0.286)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.367 (0.389)
π_{t-1}	—	—	—	0.041 (0.132)
R^2	0.298	0.313	0.318	0.306
Auxiliary regressions		Δs_t	r_t^{SP}	
$\ln \text{HYS}_{t-2}$		0.076*** (0.028)	—	
s_{t-2}		-0.182*** (0.058)	—	
$\ln[P/E10]_{t-2}$		—	-0.103*** (0.034)	
R^2		0.069	0.059	

7 Successes and Challenges

The long-run spread reproduces cleanly and the credit-beats-equity result holds ($R^2 \approx 0.38$). The main difficulties were sourcing the high-yield share and matching the paper’s Newey and West (1987) standard errors; vintage and series-stitching choices explain small differences from the published numbers.

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